

CSCC69 Operating Systems

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Why do we need operating systems?



Why learning about operating systems

- **An exciting time for building operating systems**
New hardware, smart devices, self-driving cars, data centers, etc.
Facing OS issues in performance, battery life, security, isolation
- **Pervasive principles for systems in general**
Caching, concurrency, memory management, I/O, protection
- **Understand what you use**
System software tends to be mysterious
Understanding OS makes you a more effective programmer
- **Complex software systems**
Many of you will go on to work on large software projects
OSes serve as examples of an evolution of complex systems

CSCC69

- An introductory course on Operating Systems' design principles
 - A hands-on experience building an OS
- ➡ Theory and practice goes hand-in-hand

New trends in OS

same same but different

Different architectures but, in the end, the same concepts defined in the 70s

But there are some new trends

- Multicore
- Energy efficiency (mobile and IoT devices)
- Virtualization (cloud computing)

This is a tough course ...

- Lots of things to read, materials to digest before being able to produce something that looks like the solution
- Half-way through implementing your solution, you will need to refactor it
- In the end, it will likely be one of the most challenging and significant piece of code that you have ever written

... but you will gain valuable learning and experience

Course Work

Pintos

- Developed in 2005 for Stanford's CS 140 OS class and used by many universities since then
- Written in C, built for x86 hardware
- ✓ Can run on a real machine!

Project Setup

Execute concurrent programs

- that runs on Pintos
- that runs on Bochs/Qemu emulators
- that runs on Linux
- that runs inside a Docker container
- that runs on your OS
- that runs on your personal computer



"Turtle all the way down" - Wikipedia

4 Projects

Project 1	Threads	Individual/Group	challenging
Project 2	User Programs	Individual/Group	challenging
Project 3	Virtual Memory	Individual/Group	very challenging
Project 4	File System	Individual/Group	very challenging

You can work in a group . . .

. . . but it is not a team project !

- There is no isolated tasks that can be assigned to different individuals
- ➡ All individuals must design and implement the whole solution to fully grasp the underlying concepts (that are in the details)
- ✓ Work alone first, confront your solution and come up with the best one

[disclaimer] Working as a group

- Each individual is responsible to make the whole project work
- Each individual is responsible for the work submitted as a team

Projects deliverables and grading

1. Automated tests

- All tests are given so you immediately know how well your solution performs
- You either pass a test case or fail, there is no partial credit

2. Design document

- Answer important questions related to your design for a lab

3. Coding style

- Code must be easy to read and follow coding style guidelines
- The TA will conduct a code review

The red line between collaboration and plagiarism

✓ Collaboration is allowed

- you can explain a concept to someone in another group
- you can discuss algorithms/testing strategies with other groups

⦿ Plagiarism is not allowed

- you cannot discuss specific implementation details
- you cannot look at other people's solutions, including solutions online (e.g. GitHub)
- you cannot publish your own solution online (even after the course term)

✦ Looking for materials is tolerated

- you can look for snippets of code online as long as this piece of code does not directly answer your a specific problem of the assignment
- if you copy more than 5 lines of code, put the source url as a comment in your code
- if you use more than 25 lines, do not copy it

AI Policy

⦿ You **cannot use AI** to

- Generate code that is part of the assignment solution
- Produce design-document answers

✓ You **can use AI** to

- Ask high-level conceptual questions about operating systems
- Ask questions to better understand the starter code
- Ask for feedback about your solution and your code

➡ There is an AGENTS .md file in the root directory of the project to instruct AI tools what to do and not do

You are not allowed to delete or alter this file

Late policies

Each group will have **4 days grace period** that can spread into 4 projects for interview, attending conference, errands, and so on, no questions asked

➔ **use it wisely, no more late days will be granted regardless the reason**

Let's look at the syllabus

<https://thierrysans.me/CSCC69/>

How to succeed in the course

- Stay on top of lecture materials and readings
- Start working on projects from day 1